

Let $\Sigma = \{a, b\}$. Now consider the following languages over Σ .

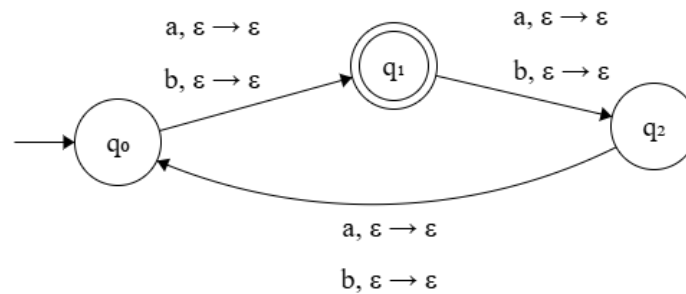
$L_1 = \{w : |w| \text{ is one more than multiple of three}\}$

$L_2 = \{a^n b^m \text{ where } 0 \leq n \leq m\}$

$L_3 = \{a^p b^q a^r b^s \text{ where } p, q, r, s \geq 1 \text{ and } p+q = r+s\}$

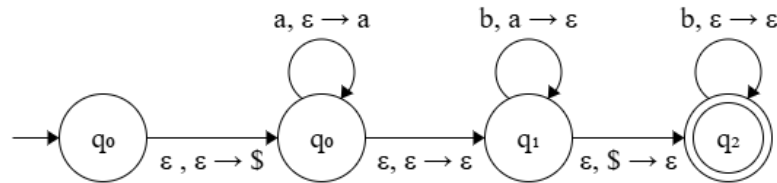
a) Construct a Pushdown Automaton for the language L_1 . [2 Points]

Ans:



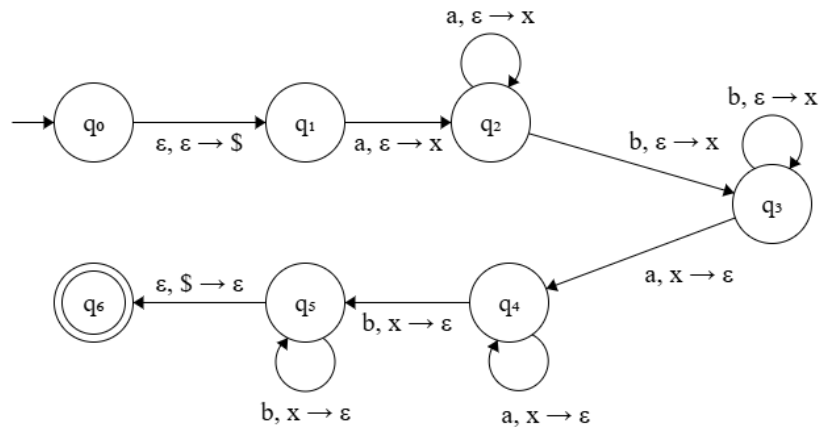
b) Construct a Pushdown Automaton for the language L_2 . [4 Points]

Ans:



c) Construct a Pushdown Automaton for the language L_3 . [4 Points]

Ans:



Let $\Sigma = \{a, b\}$. Now consider the following languages over Σ .

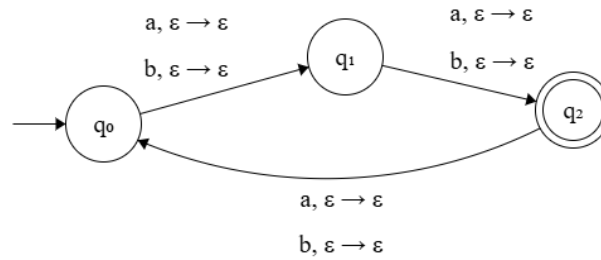
$L_1 = \{w : |w| \text{ is two more than multiple of three}\}$

$L_2 = \{b^n a^m \text{ where } 0 \leq n \leq m\}$

$L_3 = \{b^p a^q b^r a^s \text{ where } p, q, r, s \geq 1 \text{ and } p+q = r+s\}$

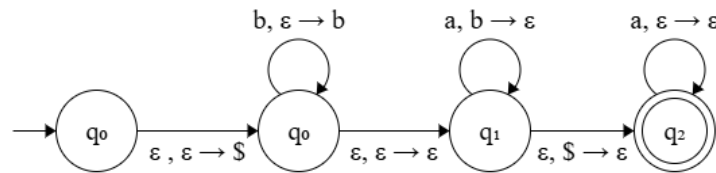
a) Construct a Pushdown Automaton for the language L_1 . [2 Points]

Ans:



b) Construct a Pushdown Automaton for the language L_2 . [4 Points]

Ans:



c) Construct a Pushdown Automaton for the language L_3 . [4 Points]

Ans:

